FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 4.2: INTRODUCTION TO SPACE PHYSICS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — STUDENT INSTRUCTIONS

Driving Question: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?

You have spent 12 lessons investigating the origin, structure, and evolution of the universe — from the Big Bang to the formation of stars, planets, and our Solar System, and from the motions of celestial bodies to how those motions shape life here on Earth. Now it is time to bring all of that evidence and reasoning together into one well-structured Final Explanation.

Your task is to write a comprehensive scientific explanation that fully answers the driving question above. Think of this as your chance to explain to a curious fellow student — perhaps one who was standing with you on the shores of Lake Victoria on a clear night — exactly what they are seeing, where it all came from, and how scientists figured it out without ever physically travelling to those distant stars or galaxies.

WHAT TO DO:

- Work through each section below in order. Each section has a guiding prompt and an exemplar sentence to help you get started.
- Use specific scientific vocabulary from the unit (e.g., Big Bang, cosmic microwave background radiation, redshift, nebula, nuclear fusion, light-year, gravitational force, electromagnetic spectrum).
- Connect your ideas to real evidence — not just definitions. Ask yourself: "How do we KNOW this?"
- Reference the anchoring phenomenon: keep coming back to what you actually see in the Nairobi night sky and explain it using what you have learned.
- Aim for depth, not length. One well-supported claim is stronger than three vague statements.
- You may use diagrams, timelines, or labeled sketches to support your written explanation.

ASSESSMENT REMINDER: Your teacher will assess your Final Explanation using the rubric attached at the end of this document. Read the rubric BEFORE you begin writing so you know exactly what "excellent" looks like.

Section 1: The Origin of the Universe — The Big Bang	
Explain how and when the universe began, according to the Big Bang theory. What was the universe like in its earliest moments? What happened in the first few minutes, and how did the first matter form? Make sure you explain WHY scientists accept this theory — what specific evidence supports it? Connect your answer to the anchoring phenomenon: why does the night sky above Nairobi look static and unchanging, yet science tells us it all exploded into existence 13.8 billion years ago?	Approximately 13.8 billion years ago, all the matter, energy, space, and even time that make up the universe began expanding rapidly from an extremely hot, extremely dense state — an event scientists call the Big Bang. In the first fractions of a second, temperatures were so high that matter and energy were interchangeable. Within the first three minutes, protons and neutrons combined to form the nuclei of the lightest elements — mainly hydrogen and helium — in a process called Big Bang nucleosynthesis. The universe then continued to expand and cool for hundreds of thousands of years before electrons joined nuclei to form neutral atoms. At that moment, light was finally able to travel freely through space. We can still detect that ancient light today as the Cosmic Microwave Background (CMB) radiation — a faint glow of microwave energy arriving from every direction in the sky, first detected in 1965. The CMB is one of the strongest pieces of evidence that the Big Bang occurred. Another key piece of evidence is redshift: when we observe the light from distant galaxies using telescopes, the wavelengths are stretched toward the red end of the electromagnetic spectrum, telling us those galaxies are moving away from us. The farther a galaxy is, the faster it recedes — exactly what we would expect if space itself is expanding from a single origin point. So when you stand on the shores of Lake Victoria and the night sky looks permanent and still, you are actually looking across billions of light-years at the stretched, cooled remnant of an ancient explosion — a universe still expanding at this very moment.

Section 2: The Formation of Stars, Galaxies, and Our Solar System	
Explain how the first stars and galaxies formed after the Big Bang, and then describe specifically how our own Solar System — including the Sun, Earth, and other planets — came into existence. What role did gravity and nuclear fusion play? Use the concept of a nebula in your answer. How does the formation story of our Solar System explain what we see when we look at the Moon, Venus, or Jupiter from Nairobi?	After the Big Bang, the universe was filled mostly with hydrogen and helium gas. Over millions of years, gravity — the attractive force between all objects that have mass — caused this gas to clump together into vast clouds called nebulae. As a nebula collapsed under its own gravity, material at its centre became denser and hotter. When the core temperature reached approximately 10 million degrees Celsius, nuclear fusion ignited: hydrogen nuclei fused together to form helium, releasing enormous amounts of energy in the form of light and heat. A star was born. The first stars were much larger and shorter-lived than our Sun. When they died in massive explosions called supernovae, they scattered heavier elements — carbon, oxygen, iron — into space. These elements became the raw materials for later stars and, crucially, for rocky planets like Earth. Our own Solar System formed approximately 4.6 billion years ago from a rotating cloud of gas and dust — a solar nebula. Gravity pulled most of the material into the centre, forming the Sun. The remaining material flattened into a spinning disc, and smaller clumps within that disc collided and stuck together over millions of years, eventually building the planets. This is why all the planets orbit the Sun in roughly the same plane, and in the same direction. The Moon, Venus, and Jupiter — visible from Nairobi on many clear nights — are all products of this same formation process, each shaped by its distance from the Sun, its size, and the materials available in its region of the solar nebula.

Section 3: Motions of Celestial Bodies and Their Effects on Life on Earth	
Describe the key motions of Earth and other bodies in our Solar System (rotation, revolution, the Moon's orbit).	Earth moves in two main ways: it rotates on its own axis once every 24 hours, and it revolves around the Sun once every 365.25 days. Earth's rotation gives us the cycle of day and night — as Nairobi faces the Sun, it experiences daylight; as it rotates away, night falls and the stars become visible. Earth's axis is tilted at approximately 23.5° relative to its orbital plane. This tilt is the reason we experience seasons. When the Southern Hemisphere tilts toward the Sun, Kenya — sitting close to the equator — experiences

Explain how these motions directly affect life on Earth — think about day and night, seasons, tides, and the calendar. Use specific Kenyan examples: how do the seasons affect farming in the Rift Valley? How do tides affect fishing communities on Lake Victoria or along the Mombasa coast? Why does the Moon appear large and bright in the Nairobi sky, and what does it actually do to Earth?	slightly more direct sunlight and its long rains season. Farmers in the Rift Valley time the planting of maize and beans around these seasonal rainfall patterns, which are directly driven by Earth's orbital position. The Moon orbits Earth roughly every 27.3 days. Its gravitational pull on Earth's oceans creates tides — the regular rise and fall of sea level. Fishing communities along the Mombasa coast and on the shores of Lake Victoria learn to read the tidal and lunar cycle, timing their fishing trips around periods of high and low water. The Moon's bright appearance in the Nairobi sky is not because it produces its own light — it reflects sunlight. Its phases, from new moon to full moon, occur because we see different portions of the Moon's sunlit face as it orbits Earth. These phases have been used by Kenyan communities for centuries to track time, plan agriculture, and navigate at night.

Section 4: How Scientists Gather Evidence About the Universe	
This is one of the most important parts of your explanation. Scientists have never physically travelled to distant stars or galaxies — yet they make detailed, confident claims about events that happened billions of years ago and objects billions of light-years away. How is this possible? Explain at least THREE specific methods or tools scientists use to gather evidence about the universe. For each one, explain what kind of information it provides and why it is reliable. Connect this back to the driving question: how do we KNOW what the night sky above Nairobi truly is?	One of the most remarkable things about modern astronomy is that scientists can learn an enormous amount about the universe without ever leaving Earth. They do this by collecting and analysing different forms of electromagnetic radiation — the family of waves that includes visible light, radio waves, infrared, ultraviolet, X-rays, and gamma rays. Each type carries different information about the objects that emit or absorb it. First, telescopes — both ground-based and space-based — collect visible light and other radiation from distant objects. By splitting starlight through a prism or diffraction grating (a technique called spectroscopy), scientists produce a spectrum — a rainbow of colours crossed by dark lines. Each element absorbs light at specific wavelengths, leaving a unique 'fingerprint.' This tells scientists exactly which elements a star is made of, even one thousands of light-years away. Second, redshift measurement allows scientists to determine how fast distant galaxies are moving away from us. When a light source moves away from an observer, its wavelengths appear stretched — shifted toward red. The greater the redshift, the faster the galaxy is receding. Belgian scientist Georges Lemaître and American astronomer Edwin Hubble used this data in the 1920s to show that the universe is expanding, which implies it must have been smaller — and hotter — in the past. Third, the Cosmic Microwave Background (CMB) radiation provides a 'snapshot' of the universe when it was only about 380,000 years old. Detected by satellites such as COBE and Planck, the CMB's near-perfect uniformity and slight temperature variations match the predictions of the Big Bang model with extraordinary precision. Together, these methods transform what looks like a quiet, permanent night sky above Nairobi into a richly readable archive of cosmic history — every photon of light carries a message across time.

Section 5: Synthesising Your Explanation — Returning to the Night Sky Above Nairobi	
Now bring everything together. Return to the anchoring phenomenon: a student	When you stand in the Rift Valley on a clear night and look up, you are not seeing a static, eternal sky. You are seeing the 13.8-billion-year story of the universe written in light. Every star you can see is a nuclear furnace — a ball of hydrogen and helium undergoing fusion, just as our Sun does — formed when gravity pulled ancient clouds of gas together billions of years ago. The

standing away from Nairobi's light pollution, perhaps in the Rift Valley or on the shores of Lake Victoria, gazes up at a sky full of stars, the Milky Way, the Moon, and perhaps Venus or Jupiter. Using everything you have learned across all 12 lessons, write a unified explanation of what that student is truly seeing. Address both parts of the driving question directly: (1) How did everything in that sky come to exist? (2) How do scientists gather evidence about a universe they cannot directly touch or travel to? Reflect on how your thinking has changed since the start of the unit.	faint, milky band stretching across the sky is our own galaxy, the Milky Way: a spiral collection of over 200 billion stars, dust, and gas, of which our Sun is just one ordinary member located about 26,000 light-years from the centre. The Moon hanging bright and large is a rocky world that formed from debris after a Mars-sized object collided with the early Earth approximately 4.5 billion years ago — and its gravity gently pulls at Earth's oceans, creating the tides that fishermen along the Kenyan coast have depended on for generations. The planet Venus, blazing near the horizon, is a rocky world that formed from the same solar nebula as Earth, but whose runaway greenhouse effect turned it into a hellish environment — a warning about what happens when a planet's atmosphere traps too much heat. None of this knowledge came from travelling to these objects. It came from carefully analysing the light they send us — using spectroscopy to read their chemical composition, measuring redshift to track their motion, and detecting ancient radiation like the CMB to reconstruct the universe's earliest moments. Scientists like Wanjiru Kamau-Rutenberg, who champions science education across Africa, remind us that the tools of inquiry are available to everyone. The night sky above Nairobi is not a mystery to be accepted in silence — it is an invitation to ask questions, gather evidence, and build models that bring us closer to understanding our place in the cosmos. At the start of this unit, the sky may have seemed unchanging and unknowable. Now you understand that it is a dynamic, evidence-rich record of cosmic history — and that science gives you the methods to read it.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy & Use of Key Concepts	All scientific claims are accurate and precisely stated. Key vocabulary (Big Bang, CMB, redshift, nuclear fusion, nebula, gravitational force, electromagnetic spectrum, etc.) is used correctly and consistently throughout. The student demonstrates deep understanding of the Big Bang theory, stellar and solar system formation, and celestial motions with no significant errors.	Most scientific claims are accurate. Key vocabulary is used correctly in most cases, with minor imprecision. The student shows solid understanding of core concepts with only small errors that do not affect the overall quality of the explanation.	Some scientific claims are inaccurate or incomplete. Key vocabulary is used inconsistently or with noticeable errors. The student shows partial understanding of core concepts, and misconceptions are present that affect the quality of the explanation.
Use of Evidence to Support Claims (NGSS SEP 6: Constructing Explanations)	The student identifies and explains at least three distinct, specific pieces of scientific evidence (e.g., CMB radiation, redshift data, spectroscopy, fossil record of craters, tidal patterns). Each piece of evidence is clearly linked to the claim it supports. The student explains WHY each piece of evidence is reliable and what it tells us about the universe.	The student identifies two or three pieces of scientific evidence and links them to relevant claims. Explanations of why the evidence is reliable are present but may lack depth in one or two instances.	The student mentions evidence but does not clearly link it to specific claims, or relies on only one type of evidence. Explanations of reliability are absent or superficial. The student tends to state facts without explaining how we know them.

Connection to the Anchoring Phenomenon & Kenyan Context	The student consistently and meaningfully returns to the Nairobi/Kenyan night sky phenomenon throughout all sections. Kenyan contexts (Rift Valley, Lake Victoria, Mombasa coast, local farming, fishing, or Kenyan scientists) are integrated naturally and accurately to ground abstract concepts in real, familiar experience. The final synthesis section compellingly unifies the phenomenon with the explanation.	The student references the anchoring phenomenon in most sections and includes at least two relevant Kenyan contexts. The connection between the phenomenon and the science is generally clear, though some sections feel disconnected from the Kenyan setting.	The student makes limited or surface-level reference to the anchoring phenomenon (e.g., a single mention in the introduction). Kenyan contexts are absent or appear forced and inaccurate. The explanation could apply to any generic context rather than specifically to Nairobi or Kenya.
Coherence, Structure & Scientific Communication	The explanation is logically organised, with clear progression from the Big Bang through to the present-day night sky. Each section flows naturally into the next, building a cumulative argument. Ideas are explained clearly in the student's own words. Any diagrams, timelines, or sketches are labelled, accurate, and meaningfully enhance the written explanation.	The explanation is generally well-organised and most sections connect logically. Writing is mostly clear with some awkward phrasing. Diagrams or sketches, if included, are mostly accurate and add some value to the explanation.	The explanation lacks a clear logical structure. Sections feel disconnected or repetitive. Writing is difficult to follow in places. Ideas appear to be listed rather than explained. Diagrams, if included, are unlabelled or inaccurate.
Addressing Both Parts of the Driving Question	The student directly and fully addresses both parts of the driving question: (1) the cosmic origin and evolution of everything visible in the night sky, AND (2) how scientists gather evidence about a universe they cannot physically access. Both parts receive balanced, thorough treatment, and the final synthesis section explicitly unifies them in a compelling conclusion.	The student addresses both parts of the driving question, but one part is treated more thoroughly than the other. The final synthesis attempts to unify both but may miss some key ideas. The overall response clearly responds to the driving question.	The student addresses only one part of the driving question, or treats both parts so briefly that the driving question is not meaningfully answered. The final synthesis is missing, very brief, or does not connect the two parts of the question.